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LOW POWER WIDE AREA NETWORK (LPWAN)

INTERNAL GUIDE
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INTRODUCTION

- The rapid growth of the Internet of Things (IoT) requires communication technologies that support long-range connectivity.
- Traditional networks consume high power and are not suitable for battery-powered IoT devices.
- LPWAN is designed to provide long-distance communication with low power consumption.

WHAT IS LPWAN?

- LPWAN (Low Power Wide Area Network) is a wireless communication technology.
- It connects low-power devices over long distances
- Mainly used for IoT applications.
- Designed to support devices with low data rate and long battery life.

KEY FEATURES OF LPWAN

- Long range communication (up to 10–15 km in rural areas).
- Low power consumption (battery life up to 10 years).
- Low data rate transmission.
- Low device cost.
- Supports massive number of IoT devices.

LPWAN ARCHITECTURE

- Components include:
- End Devices (Sensors/IoT devices)
- Gateway/Base Station
- Network Server
- Application Server
- Data flow : Device → Gateway → Network Server → Application

6 POPULAR LPWAN TECHNOLOGIES

- Examples of LPWAN technologies:

•LoRaWAN

•Sigfox

•NB-IoT

•LTE-M

Each technology differs in data rate, coverage, and cost.

APPLICATIONS

- Smart cities
- Smart agriculture
- Environmental monitoring
- Smart metering (water, electricity)
- Asset tracking
- Industrial automation

ADVANTAGES

- Very low power consumption
- Wide coverage area
- Low operational cost
- Supports large-scale IoT deployments

DISADVANTAGES

- Low data rate
- High latency
- Not suitable for real-time or high bandwidth applications

FUTURE SCOPE

- Rapid growth in IoT ecosystem
- Integration with 5G networks
- Expansion in smart city infrastructure
- Improved security and scalability

CONCLUSION

- LPWAN plays a key role in enabling large-scale IoT connectivity.
- It offers long-range communication with minimal power consumption.
- It is an essential technology for future smart applications.

REFERENCES

- Research papers on LPWAN
- IoT networking textbooks
- IEEE publications
- Official documentation of LoRaWAN and NB-IoT